Swarm Robots for Local Mars Resource Localization and Mining

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Abstract

This project proposes the development of an autonomous swarm-based robotic framework for distributed resource localization on the Martian surface. The concept leverages many relatively low cost robotic agents that cooperate to map terrain, detect scientifically relevant signatures, and report potential resource sites. Unlike single-rover architectures this swarm approach improves coverage, resistance to failure, and adaptability to uncertain terrain and degraded communication. The work plan focuses on multi-robot autonomy, sensing trade studies, SLAM algorithm evaluation, and simulation-based testing using platforms such as SpaceROS, NVIDIA Isaac Sim, ARGoS, and OmniLRS. Through modeling, analysis and subsystem design, this effort advances technology supporting NASA's goals in Robotics and autonomous systems under TX04, particularly system-level autonomy (TX04.5).

Technology Merit and Work Plan

Research into autonomous systems has been undergone at both industry and federal levels. The technology we seek to adapt towards space exploration has been developed extensively in the past few decades. This proposal seeks to repurpose proven technologies in environments we can simulate on Earth at NASA facilities.

Concept Overview

Our concept deploys a fleet of small, low cost robots working cooperatively to survey wide areas of the martian surface. Each robot performs local sensing, navigation and mapping, while the swarm collectively fuses data to identify and validate resource signatures. The distributed nature of the swarm allows resilient operation even when individual robots fail or communications degrade. This architecture addresses NASA's long standing challenge of limited surface coverage, high system risk, and strong evidence on a single rover's mobility and sensing capabilities. Swarm robotics are based on natural hive systems observed in species like honeybees, where many workers give the central system greater adaptability and efficiency.

Existing Limitations

Guidance, navigation, and control (GNC) is the process used to design a system that controls movement of a vehicle. This is especially important for autonomous robotic swarms as it provides the foundation for mapping and movement in a region to be explored. Simultaneous localization and mapping (SLAM) is one method used for autonomous vehicles that achieves this. To understand the present limitations of swarm robotics, it is integral to understand the different implementations of SLAM.

There are three broad categories of SLAM, two of which will be discussed concerning their limitations. Visual SLAM (or vSLAM) uses images acquired from cameras and other image sensors [1]. Cameras and image sensors used by the unmanned ground vehicles may be obstructed by the Martian environment. If we compensate for this by using images and videos of previously explored regions on Mars, we are limited to the locations of 7 landers and 4 rover sites [2].

Light detection and ranging (lidar) SLAM utilizes a laser or distance sensor. Compared to cameras and other visual sensors, lasers are significantly more precise in comparison to mapping out terrain with complex geometries. These geometries are then able to be translated into 2D and 3D point clouds which are used by estimation methods that will be discussed further into the writing [1]. However, limitations arise when we consider the general environment of Mars. Most lidar systems don't operate well in temperatures below -10 degrees Celsius due to the nature of semiconductors. This means that the reliability of lidar SLAM and related estimation methods may come into question [3]. Additionally, the radiation experienced by robotic systems on Mars will also affect lidar as background noise generated can easily cause false alarms in the receiver, which affects the effective detection of the system [4].

State of the Art Assessment

Swarm robotics is not a new field, and it is growing constantly. The uses of swarms on Earth range from entertainment-based quadcopter light shows to applied science like environmental monitoring. As of today, swarm robots used for resource localization and extraction in space are infantile in research and deployment.

Currently there is a non-swarm based solution for localization and mining called Regolith Advanced Surface Systems Operations Robot (RASSOR). RASSOR is a teleoperated rover capable of excavating low gravity bodies. It has a reduced weight from rovers such as Opportunity and is foldable which reduces payload dimensions. RASSOR is capable of autonomy but only for specific tasks.[10]

There are many studies and papers on swarm resource collection but not many realized robots or simulations.[7, 8, 9] One swarm based project, Lunarminer, seeks to apply swarm robotics to lunar-based resourcing of water-ice in permanently shadowed craters. In their paper researchers claim a possible reduction of extraction time by 40% and a 31% reduction in energy consumption. These reductions are paired with cost reductions mainly through eliminating the need to send water to the moon (as current estimates place costs at 35 to 70 thousand dollars per kilogram of water) [11]. Their proposed system is estimated to be able to gather 181 liters of water per day, enough for 18 astronauts.These improvements are overshadowed by the project's robot failure rate of 20%, but the core technology behind swarms are theoretically more fault tolerant, adaptable, and scalable than larger rovers with the same goals, and also fully autonomous.[11]

In-situ Resource Utilization (ISRU) in regards to swarm robotics is well studied from an algorithms, testing procedures, and use-case perspective–but these need to be fully realized to be useful. Swarm Robotics for extra terrestrial use is a promising-but-still-emerging technology, and more research and development is required before commitment to its use in space.

Technology Challenges and Risks

Research has been conducted to ascertain how effective various Simultaneous Localization and Mapping (SLAM) algorithms can be implemented in Unmanned Ground Vehicles–each with its own disadvantages. For example, the Extended Kalman Filter algorithm has trouble dealing with large maps as issues can arise. The FastSLAM algorithm depends on particles that are associated with a possible pose for the vehicle and landmarks. This leads to degeneracy as the process when sampling the proposal distribution requires the history of the particle. However this disadvantage is mitigated by FastSLAM 2.0, which decreases the rate of degeneracy. Finally the Graph SLAM algorithm has a high computational cost compared to the other algorithms due to the increased number of vehicle poses and landmarks within a small area. This disadvantage is then amplified the larger the map is [12].

Environmental conditions also pose a challenge to the reliability of SLAM algorithms. For instance, dusty environments remain as a limitation to be further researched before this technology can be applied [5.] Thus, applications of SLAM algorithms would need to be adjusted for navigating in environments that may hinder performance. For deployment on Mars and in other harsh environments, new care and consideration needs to be taken when developing the swarm, as current implementations of the technology are terrestrial and do not pose as much of a threat.

Goal Performance Metrics

Goal performance metrics for the GNC Design subsystem can be a verification system between drones. Although prone to inaccuracies due to limitations discussed earlier in the writing, bearings and distances cannot be measured accurately through other means such as with the Mars Reconnaissance Orbiter. On-the-ground verification would be suited best to ensure that position, velocity, drift are within acceptable parameters as is described in Attitude and Position Estimation on the Mars Exploration Rovers [18].

Evaluations of the swarm effectiveness should be the cost to performance ratio (performance being materials collected and work done) compared to previous rovers deployed on Mars, as well as longevity.

Workflow Overview

Since the field is still barren of realized space swarm robots for resource localization and extraction, the ability to follow through with the build is critical. To realize the technology a considerable amount of effort has been made in preparation of *how* to realize the project. We will have 2 weekly meetings, one for progress updates and one for design choices. The progress meetings will also serve the purpose of making design difficulties known to aid any such subsystem. The design meeting will be for overall project goals, maintaining compliance with the project requirements and specifications, and the integration effort for the subsystems. Subsystem teams will utilize the software management tool to stay current on their design implementations as well as others. Our project requires the following subsystems:

- Hardware Design
    - Robot Mechanical Design (CAD Models for simulators)
    - Robot Embedded Sensors (What sensors/devices we need to model)

- Software Design
    - Swarm Algorithm Design
    - Machine Learning and AI Design
    - Sensor Firmware Design
    - Mobility Firmware Design
    - Control Systems Design
    - GNC Design

This simulation design will include the use of the Robot Operating System 2 (ROS2), in the form of SpaceROS[12], the OmniLRS simulation project[14], the multi-agent_sim project[15], NVIDIA Isaac Sim [16], and ARGoS[17]. Each simulator will serve different purposes in finalizing the software for the swarm. The software will be managed using Git to allow each project member ease of access in the collaborative nature of the project.

Project Management Approach

This project follows a structured systems engineering management approach that aligns team roles, responsibilities and development phases to support a 12-month technology development effort that will be appropriate for NPWEE. Since there is no physical testing during our proposal period, we will focus on modeling, analysis, concept design and prototype planning that is consistent with NPWEE expectations.

Management Structure

The team's organization is based on the NPWEE Team 5 org chart that clearly defines leadership roles, technical roles as well as the proposal responsibilities. The leadership roles are broken into four categories: Principal Investigator (PI), Project Manager (PM), Lead Engineer/Deputy PM and Admin Lead. We then have our five technical/support roles: Lead Scientist, Software Engineers, Robot CAD Modeling & Hardware Research, Budget & Basis of estimate (BOE) lead and the Visualization & Data Analysis role. We have some roles that will be filled once we receive the funding to continue this proposal which would be the Simulation Specialist, Mechanical CAD Engineer, RF Engineer and the GNC Engineer.

12-Month Project Schedule

Our project will follow a structured five-phase development timeline over a 12-month base period. We will incorporate about a 15-20% schedule margin in the later phases to account for any time needed in terms of modeling iteration, SME cycle review, documentation or any delays. This will ensure execution and on schedule planning.

<u>Phase 1:</u> Requirements & Mission Definition (Months 0-2)
Activities: Environmental constraints, national needs, system requirements.
Deliverables: Requirements Document & Initial Architecture

<u>Phase 2:</u> Concept Development & Modeling (Months 2-5)
Activities: Concept architecture, autonomy behaviors, sensing & communication trade studies, early modeling.
Deliverable: Concept Architecture Package & Modeling Results.

<u>Phase 3:</u> Technical Risk Assessment & Prototype Planning (Months 5-8 with 3 weeks schedule margin)
Activities: Risk identification, feasibility analysis, mitigation planning, prototype plan outline.
Deliverable: Risk Assessment Matrix & Prototype Planning Document.

<u>Phase 4:</u> Preliminary Design & Integration Planning (Months 8-11 with 2 weeks schedule margin)
Activities: Preliminary subsystem designs, interface definitions, data/communication flows, integration plans.
Deliverable: Preliminary Design Document (PDD) & Interface Control Document (ICD).

<u>Phase 5</u>: Final Documentation & Closeout (Months 11-12 with 1 week schedule margin)
Activities: Final report assembly, modeling refinement, documentation consolidation, transition recommendations.
Deliverable: Final Report.

Resources, Skills and Basis of Estimate (BOE)

We will identify the labor resources, required skillsets and BOE for each project element. All labor hours are included even though they do not carry direct cost. Estimates are based on NPWEE guidance for time-per-task, number of cycles, experience level and documentation.

Project Manager (PM)
Skills: Systems engineering, scheduling, documentation, risk tracking.
BOE: 120- 50 hrs: Weekly PM duties + major review preparation.

Principal Investigator (PI)
Skills: Scientific oversight, technical validation.
BOE: 40-50 hrs: Monthly oversight + review participation.

Deputy PM/Lead Engineer
Skills: Engineering integration, subsystem coordination.
BOE: 40-60 hrs: Support for subsystem cycles and preliminary design review.

Lead Scientist
Skills: Geology, meteorology, environmental modeling.
BOE: 60-80 hrs: Environmental parameters, site constraints, modeling inputs.

Software Engineers
Skills: Autonomy algorithms, swarm modeling, simulations.
BOE: 150-200 hrs combined: Three modeling cycles + software trade studies.

Robot CAD Modeling & Hardware Research
Skills: Mechanical modeling, early CAD layouts, subsystem interfaces.
BOE: 40-70 hrs: Conceptual models and mechanical feasibility research.

Budget & Basis of Estimate Lead
Skills: Cost modeling, BOE development.
BOE: 60-80 hrs: Cost research, BOE creation, and revisions.

Graphics, Visualization and Data Analysis
Skills: System diagrams, graphics, data visualization.
BOE: 40-60 hrs: Architecture diagrams, system visuals, and flow charts.

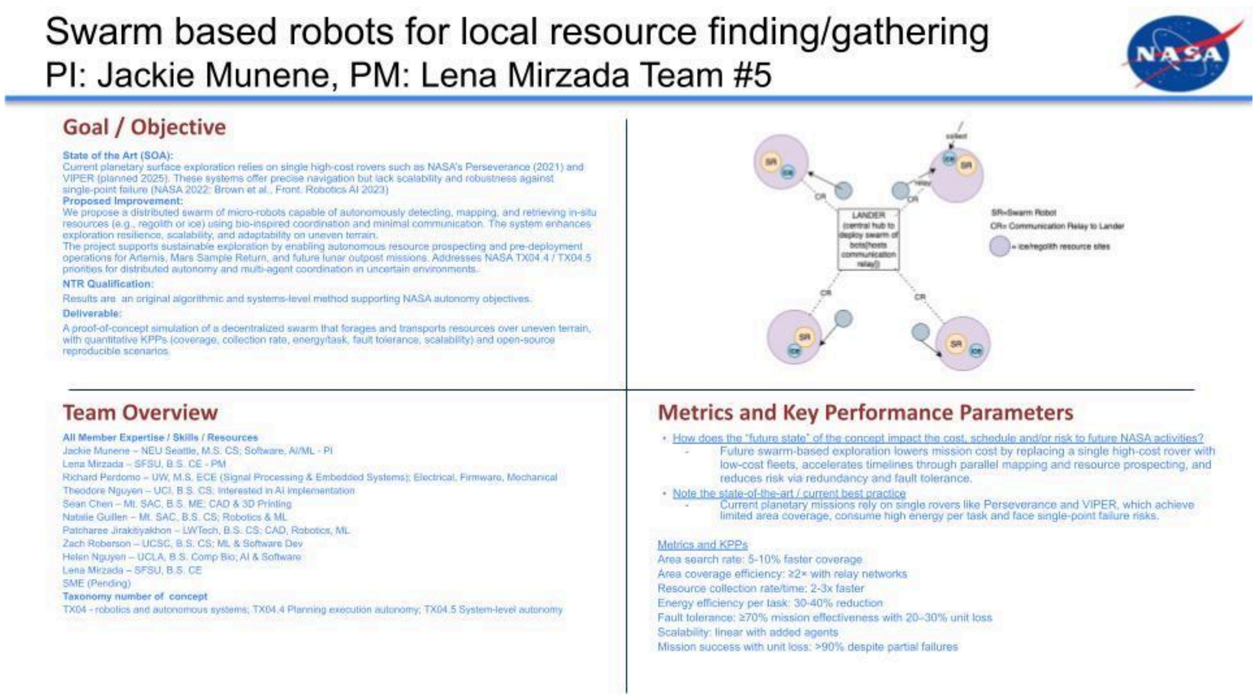
Subject Matter Expert
Skills: Robotics/autonomy domain expertise.
BOE: 20-30 hrs: Monthly reviews + milestone validation.

Teaming and Workforce Development

The team is led by PI Jackline Munene, who provides scientific oversight, requirement validation and technical guidance (~40-50 hrs). Deputy PM/Lead Engineer Richard Perdomo supports subsystem coordination, engineering integration and technical reviews (~40-60 hrs). Project Manager Lena Mirzada oversees scheduling, deliverables, documentation and systems integration, ensuring alignment with NASA and national needs (~50-60 hrs). Software Engineers Natalie Guillen and Zachary Roberson develop autonomy algorithms, multi-agent coordination methods and simulation concepts (~150-200 hrs combined). Lead Scientist Helen Nguyen contributes geology, meteorology and environmental modeling expertise (~60-80 hrs). Sean Chen supports early CAD modeling, mechanical layout development and communication subsystem research (~40-70 hrs). Theodore Nguyen prepares the budget, Basis of Estimate (BOE) and cost justification (~60-80 hrs). Patcharee Jirakittiyakhon develops system diagrams, architecture graphics and data visualizations (~40-60 hrs). David Aldana provides GNC design support (~30-50 hrs). An SME will provide technical validation and milestone review support (~20–30 hrs). The project includes skill development in autonomy, CAD, systems engineering and scientific analysis, with remaining gaps addressed through SME mentorship and cross-training.

Appendix

Appendix A. Quad chart



Appendix C. (Optional) Science traceability matrix

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